

Table of content

1. Introduction
2. Aims and Objectives
3. Dataset description
4. Exploratory Data
 - 4.1 Statistical data description
 - 4.2 Bar graph analysis
 - 4.3 Box plot analysis
5. Data preprocessing
 - 5.1 Removal of outliers
 - 5.2 Handling missing values
 - 5.3 Data splitting and normalization
6. Classification models
 - 6.1 Random forest model
 - 6.2 Support Vector Machine model
7. Clustering and Classification Analysis
 - 7.1 K-mean clustering model
 - 7.2 Random forest classifier model
 - 7.3 Making predictions and result analysis
8. Regression based models
 - 8.1 Linear regression model
 - 8.2 LSTM model
 - 8.3 LSTM as superior model
 - 8.4 Implications of eBay's auction strategy

References

List of Figure

Figure 1: Shill bidding dataset

Figure 2: eBay auction dataset

Figure 3: statistical data distribution

Figure 4: Bar graph analysis for the distribution and impact on auction outcomes

Figure 5: Box plot analysis for the distribution and impact on auction outcomes

Figure 6: Removal of outliers

Figure 7: model evaluation matrices and graphs of random forest

Figure 8: model evaluation matrices and graphs of SVM

Figure 9: Segmentation categorized bidders into three distinct classes: 0: 'Aggressive Bidders', 1: 'Passive Bidders', and 2: 'Balanced Bidders'

Figure 10: model evaluation matrices and graphs of random forest classifier

Figure 11: model evaluation graph of linear regression model

Figure 12: model evaluation graph of LSTM model

1. Introduction

The way individuals buy and sell items has been completely transformed by online marketplaces like eBay, which provide a venue for consumers to take part in auctions from the comfort of their own homes. However, dishonest practices like shill bidding have the potential to undermine the integrity of these auctions. Shill bidding refers to dishonest bidders who increase auction prices in order to falsely create the impression of competition. This erodes the confidence of sincere buyers and sellers in addition to warping market dynamics.

This paper focuses on analyzing eBay auction data to detect and reduce shill bidding activity in order to address these issues. A number of characteristics, including bidder tendencies, bidding ratios, successive outbidding, and more, are included in the dataset that is being made available. By exploring and analyzing this data, we aim to develop robust models that can predict and classify potential shill bidding, ultimately enhancing eBay's auction integrity and informing strategic business decisions.

2. Aims and Objectives

Aim

The primary aim of this report is to leverage advanced data analytics techniques to detect fraudulent bidding activities, specifically shill bidding, and to forecast auction outcomes on eBay. This will help enhance the integrity of eBay's auction platform and inform strategic business decisions.

Objectives

- The first objective is to conduct a comprehensive exploratory data analysis (EDA) on the provided shill bidding dataset to uncover underlying patterns and anomalies. This involves identifying and addressing missing values and outliers to ensure data quality and reliability. Necessary data preprocessing steps will be executed to prepare the dataset for further analysis. Additionally, key variables, including Bidder Tendency, Bidding Ratio, and Auction Bids, will be visualized to understand their distributions and potential impacts on auction outcomes.
- The second objective is to develop and implement two classification models aimed at predicting whether a bid is normal or indicative of shill bidding. The performance of these models will be evaluated and compared using accuracy and other relevant metrics.

to identify the most effective model. Following this, a clustering algorithm will be applied to segment auctions based on distinctive bidding patterns, creating meaningful classes for further analysis. These clustering results will then be used to develop a classification model, make predictions, and provide an in-depth explanation of the outcomes.

- The third objective is to conduct two regression analyses to predict final auction prices using features deemed relevant to auction outcomes. The selection of features for each regression model will be justified, ensuring they are well-founded in the context of auction dynamics. The performance of the regression models will be compared to determine which one offers better predictive accuracy. Finally, the implications of the regression findings for eBay's auction strategy will be discussed, offering insights that can help optimize pricing and bidding practices.

By meeting these objectives, this report aims to deliver actionable insights that will help eBay reinforce the integrity of its auction platform, fostering a fair and transparent marketplace for all participants.

3. Dataset description

We have two datasets that are i.e shill bidding dataset and eBay auction data . The Shill Bidding dataset is used to identify and analyze patterns related to shill bidding activities in online auctions as shown in [figure 1](#). Shill bidding is the act of placing fake bids on an auction to artificially inflate the price or desirability of an item. By analyzing features like Bidder_Tendency, Bidding_Ratio, and Successive_Outbidding, it is possible to develop models that can predict whether a given set of bids constitutes shill bidding.

The eBay Auction dataset is used to analyze various factors influencing the final price of items sold through eBay auctions as shown in [figure 2](#). By examining features such as Starting_Price, Number_of_Bidders, and Auction_Duration_Days, predictive models can be developed to estimate the Final_Price of auction items based on these features.

[Figure 1; Shill bidding dataset](#)

Record_ID	Auction_ID	Bidder_ID	Bidder_Tendency	Bidding_Ratio	Successive_Outbidding	Last_Bidding	Auction_Bids	Starting_Price_Average	Early_Bidding	Winning_Ratio	Auction_Duration	Class	
0	1	732	_***j	0.200000	0.400000	0.0	0.000028	0.0	0.993593	0.000028	0.666667	5	0
1	2	732	g***r	0.024390	0.200000	0.0	0.013123	0.0	0.993593	0.013123	0.944444	5	0
2	3	732	t***p	0.142857	0.200000	0.0	0.003042	0.0	0.993593	0.003042	1.000000	5	0
3	4	732	7***n	0.100000	0.200000	0.0	0.097477	0.0	0.993593	0.097477	1.000000	5	0
4	5	900	z***z	0.051282	0.222222	0.0	0.001318	0.0	0.000000	0.001242	0.500000	7	0

Figure 2; eBay auction dataset

	Starting_Price	Number_of_Bidders	Auction_Duration_Days	Bidder_Activity	Final_Price
0	55.33	18	2	2.450	60.73
1	71.80	7	11	0.827	94.44
2	60.67	11	9	3.244	87.54
3	54.94	6	14	2.058	67.31
4	42.94	5	11	2.057	59.19

4. Exploratory Data Analysis

4.1 Statistical data description

As seen in [figure 3](#), the Shill Bidding Dataset offers a thorough summary of bidding patterns along with a number of important statistical variables. With a mean of 0.1425 and a standard deviation of 0.1971, the bidder tendency shows that, although with significant variability, bidders typically engage in auctions featuring a small number of sellers. The Bidding Ratio exhibits a range of bidding frequencies, with a mean of 0.1277 and values spanning from 0.0118 to 1. With a mean of 0.1038 and the majority of values at 0, successive outbidding is comparatively uncommon. The mean of last bidding, which indicates significant inactivity around auction ends, is 0.4631. The large variation in auction bids (mean 0.2316) is indicative of higher bids in auctions influenced by shilling. Differential starting prices and early bidding are indicated by the Starting Price Average (mean 0.4728) and Early Bidding (mean 0.4307). The Winning Ratio (mean 0.3677) highlights that many bidders win few auctions. Auction Duration averages 4.6151 days, with notable variation. The Class variable, with a mean of 0.1068, indicates that shill bidding is relatively rare in the dataset. These statistics form a solid foundation for detecting fraudulent activities and predicting auction outcomes.

Figure 3; statistical data distribution

	Bidder_Tendency	Bidding_Ratio	Successive_Outbidding	Last_Bidding	Auction_Bids	Starting_Price_Average	Early_Bidding	Winning_Ratio	Auction_Duration	Class
count	6321.000000	6321.000000	6321.000000	6321.000000	6321.000000	6321.000000	6321.000000	6321.000000	6321.000000	6321.000000
mean	0.142541	0.127670	0.103781	0.463119	0.231606	0.472821	0.430683	0.367731	4.615093	0.106787
std	0.197084	0.131530	0.279698	0.380097	0.255252	0.489912	0.380785	0.436573	2.466629	0.308867
min	0.000000	0.011765	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	1.000000	0.000000
25%	0.027027	0.043478	0.000000	0.047928	0.000000	0.000000	0.026620	0.000000	3.000000	0.000000
50%	0.062500	0.083333	0.000000	0.440937	0.142857	0.000000	0.360104	0.000000	5.000000	0.000000
75%	0.166667	0.166667	0.000000	0.860363	0.454545	0.993593	0.826761	0.851852	7.000000	0.000000
max	1.000000	1.000000	1.000000	0.999900	0.788235	0.999935	0.999900	1.000000	10.000000	1.000000

4.2 Bar graph analysis

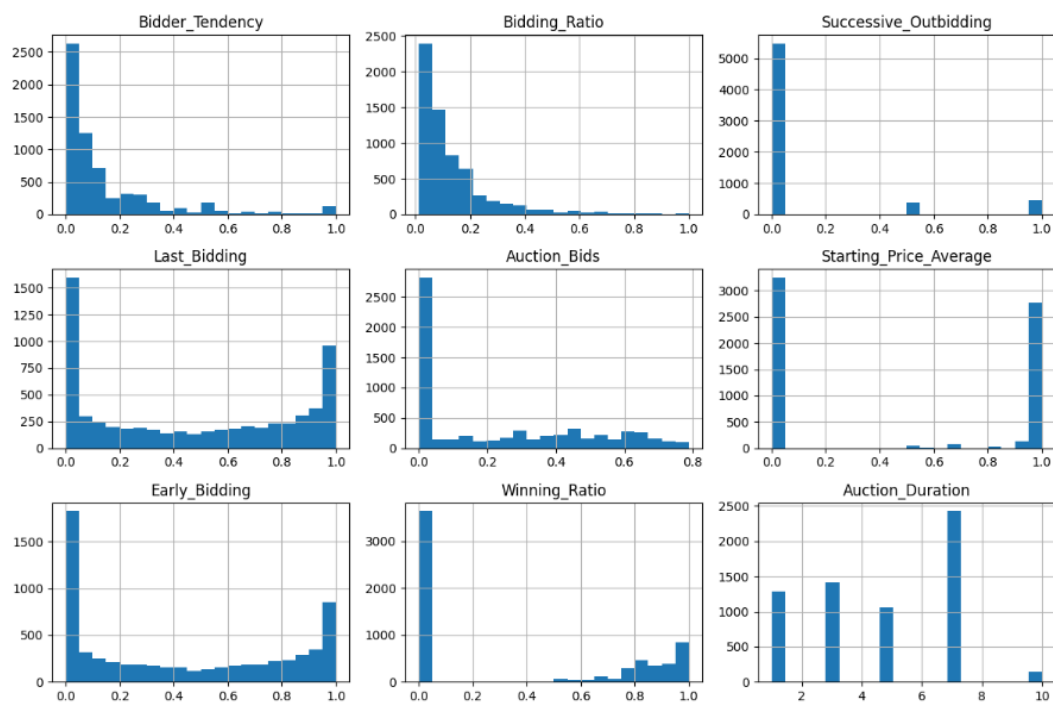
The histograms provided offer a detailed analysis of various features related to auction behavior and outcomes as seen in [figure 4](#). The Bidder_Tendency distribution is highly skewed to the right, indicating that most bidders participate infrequently, suggesting that a small number of highly active bidders dominate the auction landscape. Similarly, the Bidding_Ratio is skewed to the right, reflecting that most participants place a low proportion of total bids, hinting at conservative or strategic waiting behaviors. The Successive_Outbidding distribution shows that this behavior is rare, as most bidders do not re-enter the auction once outbid, indicating low levels of aggressive bidding.

The Last_Bidding and Early_Bidding distributions are both U-shaped, revealing two dominant strategies: early bidding and last-moment bidding, often referred to as "sniping." The Auction_Bids distribution is skewed to the right, with most auctions receiving a low to moderate number of bids, suggesting that many auctions are not highly competitive. The Starting_Price_Average also displays a U-shaped distribution, indicating that auctions tend to start at either very low or very high prices, reflecting diverse auction strategies or item types.

The Winning_Ratio is skewed to the right, showing that a small fraction of bidders win a significant proportion of auctions, likely due to superior strategies or greater resources. Lastly, the Auction_Duration distribution is relatively even with peaks at specific durations, indicating wide variability in how long auctions last, which can affect bidder engagement and final prices.

Overall, these patterns suggest that auction outcomes are influenced by varying bidding behaviors, levels of competition, and auction designs. The data reveals a mix of sniping and early engagement strategies, low competition in many auctions, and a few dominant bidders. Auction design elements such as duration and starting prices play a crucial role in shaping these outcomes. Understanding these dynamics can aid in designing more effective auctions and developing strategies for bidders to enhance their success rates.

Figure 4; Bar graph analysis for the distribution and impact on auction outcomes



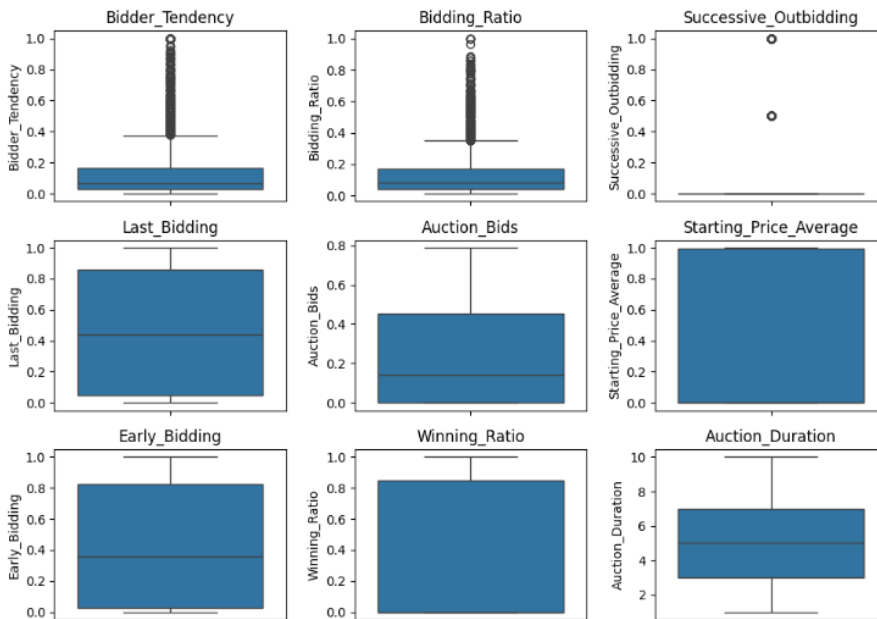
4.3 Box plot analysis

The box plots provide a comprehensive view of the distribution, central tendency, and variability of various auction-related features, highlighting key aspects of bidder behavior and auction outcomes as shown in [figure 5](#). The Bidder_Tendency data is highly concentrated towards the lower end, with a median value around 0.2 and significant outliers above 0.6. This indicates that while most bidders participate infrequently, a small number are highly active, potentially skewing auction dynamics. Similarly, the Bidding_Ratio shows a low median around 0.2 with many outliers, suggesting that most bidders place a small fraction of the total bids, though some bidders contribute disproportionately, reflecting diverse bidding strategies.

The Successive_Outbidding plot reveals a median of 0 with few significant outliers, indicating that continuous bidding after being outbid is rare, which may point to a strategy of avoiding escalating bid wars. In contrast, Last_Bidding has a median around 0.5 and a symmetric distribution, suggesting a balanced mix of early and last-moment bidding strategies. The Auction_Bids distribution shows a median around 0.2, indicating that auctions generally receive a low to moderate number of bids, reflecting limited competition in terms of bid volume.

The Starting_Price_Average exhibits a relatively uniform distribution with a median around 0.5, suggesting diverse auction strategies or item characteristics without extreme deviations. Early_Bidding also shows an even split with a median around 0.5, reinforcing the presence of mixed bidding strategies, where bidders are equally likely to place early or late bids. Lastly, the Winning_Ratio has a median around 0.2, with a broad distribution and no outliers, indicating that while most bidders have a low winning ratio, there is a wide range of success rates among participants. These patterns underscore the variability in bidding behaviors and auction strategies, which collectively shape auction outcomes.

Figure 5: Box plot analysis for the distribution and impact on auction outcomes



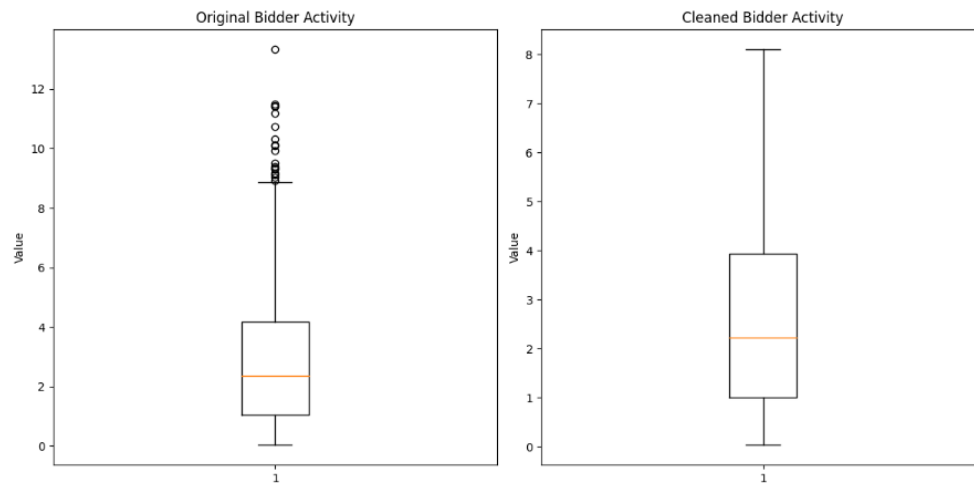
5. Data preprocessing

5.1 Removal of outliers

We concentrated on eliminating outliers from our dataset during the preprocessing stages in order to improve its quality and dependability. The effect of this step on the "Bidder Activity"

variable is depicted in the box plots above. The original data is displayed in [figure 6](#) left plot, where a number of outliers substantially above the upper quartile range, pointing to abnormally high bidder activity levels. These anomalies have the ability to distort analysis results and maybe produce false conclusions. We used outlier removal techniques to address this, and the cleaned dataset is shown in the right plot. With extreme values removed, the cleaned data now shows a more uniform distribution, which improves the portrayal of usual bidder behavior. This step is crucial for improving the performance and robustness of subsequent analytical models by ensuring that they are not unduly influenced by anomalous data points.

Figure 6; Removal of outliers



5.2 Handling missing values

Both the eBay Auction Dataset and the Shill Bidding Dataset have no missing values, according to our examination of the datasets. Both datasets have all of their columns fully populated with data, including the important variables such as Bidder Tendency, Bidding Ratio, Successive Outbidding, Last Bidding, Auction Bids, Starting Price Average, Early Bidding, Winning Ratio, Auction Duration, Class, Starting Price, Maximum Number of Bidders, Auction Duration Days, Bidder Activity, and Final Price. Our preprocessing procedures are made simpler by this completeness since it eliminates the requirement for imputation approaches and the removal of partial entries. The lack of missing values guarantees a strong and intact dataset for our further research and model training, which improves the accuracy and dependability of our findings.

5.3 Data splitting and normalization

We carefully segmented the dataset into features and target variables, normalized the features, and divided the data into training and testing sets as part of our data preprocessing to get it ready for model building. The target variable (class) was labeled as Y, and the features (bidder tendencies, bidding ratios, successive outbidding, last bids, auction bids, starting price average,

early bidding, winning ratio, and auction duration) as X. We normalized X by scaling the data to a standard range, which lessens the effect of different scales and units across variables and guarantees that each feature contributes equally to the model. Following normalization, the dataset was divided into training and testing sets, with 80% of the data going to the former and 20% to the latter. This split allows us to train our models on a substantial portion of the data while reserving a significant portion for unbiased evaluation, ensuring robust model performance and reliable validation results.

6. Classification models

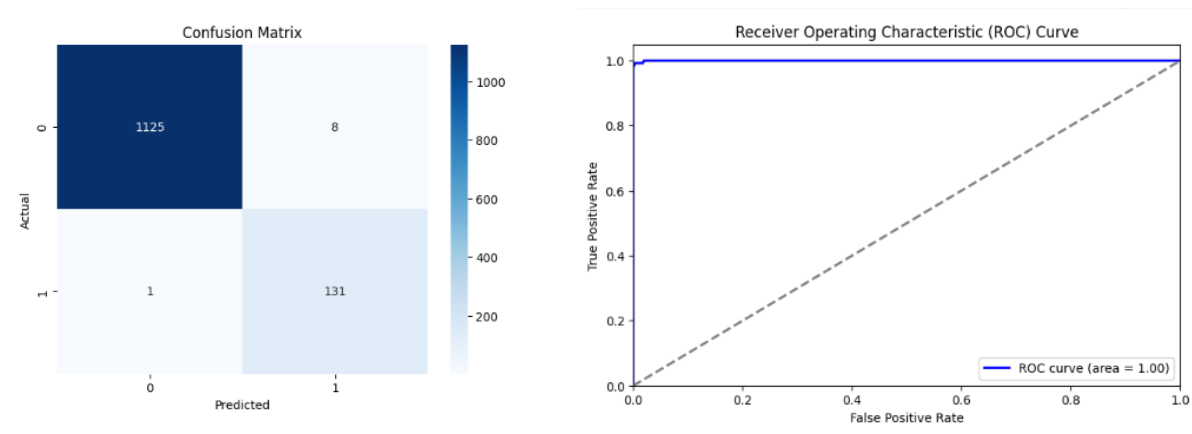
To predict whether a bid is normal or indicative of shill bidding, we implemented two classification models: Random Forest and Support Vector Machine (SVM). These models were selected for their proven effectiveness in handling classification tasks and their ability to provide robust and accurate predictions.

6.1 Random forest model

The Random Forest model demonstrated outstanding performance with an accuracy of 99.29%, precision of 99.32%, recall of 99.29%, and an F1 score of 99.30%. The detailed classification report as shown in [figure 7](#) revealed that the model performed exceptionally well in distinguishing between normal and shill bidding activities, with particularly high precision and recall rates for both classes. Specifically, it achieved a perfect precision of 1.00 for normal bids and 0.94 for shill bids, indicating its robustness in identifying fraudulent activities.

Figure 7: model evaluation matrices and graphs of random forest

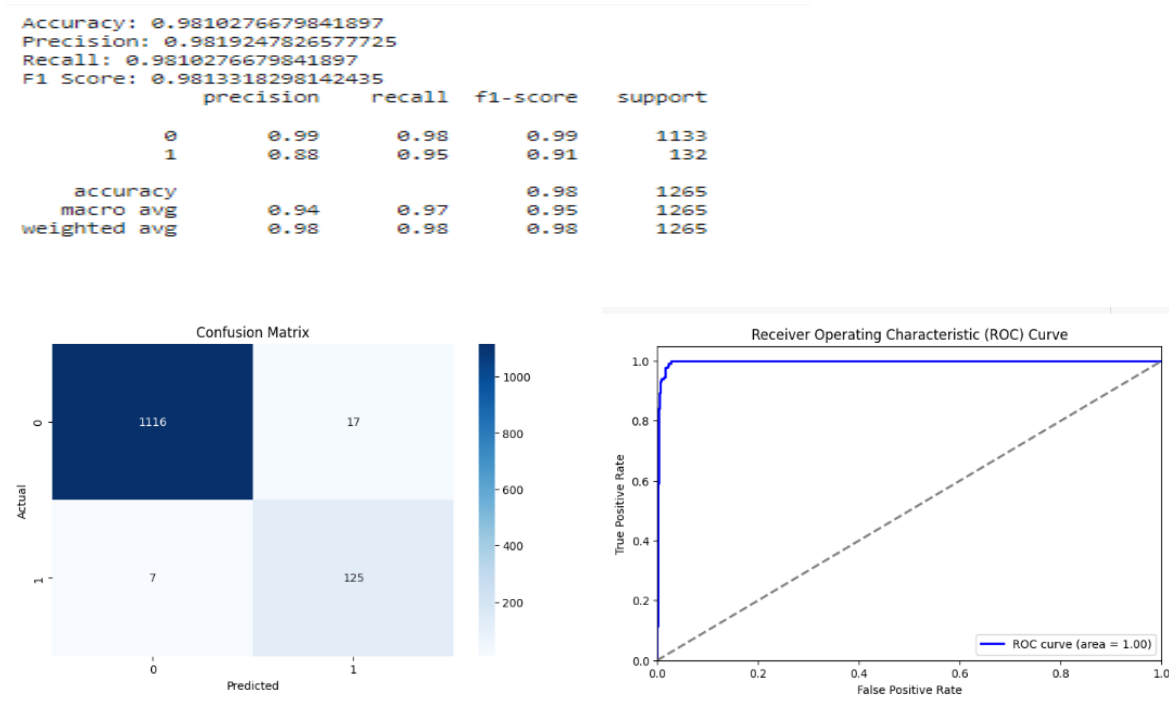
Accuracy: 0.9928853754940712					
Precision: 0.9931989415032186					
Recall: 0.9928853754940712					
F1 Score: 0.9929662386871575					
	precision	recall	f1-score	support	
0	1.00	0.99	1.00	1133	
1	0.94	0.99	0.97	132	
accuracy			0.99	1265	
macro avg	0.97	0.99	0.98	1265	
weighted avg	0.99	0.99	0.99	1265	



6.2 Support Vector Machine model

The SVM model also performed strongly, achieving an accuracy of 98.10%, precision of 98.19%, recall of 98.10%, and an F1 score of 98.13%. The classification report as shown in [figure 8](#) showed high metrics for normal bids, with a precision of 0.99, but slightly lower precision of 0.88 for skill bids. Despite its high overall accuracy, the SVM model's performance metrics were slightly lower than those of the Random Forest model, particularly in identifying skill bidding activities.

Figure 8: model evaluation matrices and graphs of SVM



Model evaluation comparison and conclusion

The comparison of the Random Forest and SVM models reveals that the Random Forest model outperforms the SVM model across several key metrics. The Random Forest model has higher accuracy (99.29% vs. 98.10%), precision (99.32% vs. 98.19%), recall (99.29% vs. 98.10%), and F1 score (99.30% vs. 98.13%). The confusion matrix for the Random Forest model shows fewer misclassifications, with only 8 false positives and 1 false negative, compared to 17 false positives and 7 false negatives for the SVM model. Furthermore, both models have a perfect ROC area of 1.00, indicating excellent discriminatory power, but the Random Forest model achieves this with fewer errors.

The superior performance of the Random Forest model can be attributed to its ensemble learning approach. By combining the predictions of multiple decision trees, the Random Forest model can capture complex patterns and interactions in the data more effectively than a single classifier like SVM. This ensemble method helps to reduce overfitting and improve generalization to unseen data, making it particularly adept at handling the nuances of shill bidding detection.

Given the higher accuracy, precision, recall, and F1 score, as well as the fewer misclassifications observed in the confusion matrix, the Random Forest model is clearly the superior choice for predicting shill bidding. Its robustness and reliability make it an invaluable tool for enhancing the integrity of online auctions on platforms like eBay, ensuring that fraudulent bidding activities are accurately identified and mitigated.

7. Clustering and Classification Analysis

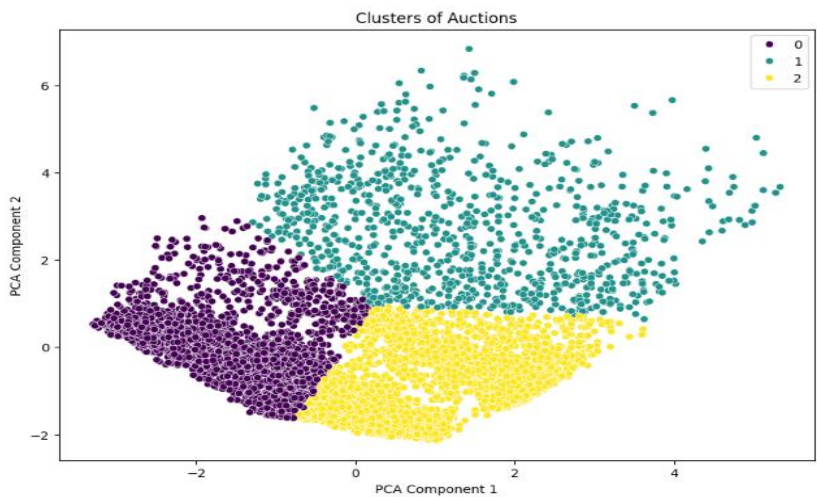
7.1 K-mean Clustering model

To further enhance the detection of shill bidding activities, I have implemented K-mean clustering algorithm to segment the auctions based on bidding patterns. This segmentation categorized bidders into three distinct classes: 'Aggressive Bidders', 'Passive Bidders', and 'Balanced Bidders'. The clustering is based on key bidding behaviors and metrics from the dataset, such as Bidder Tendency, Bidding Ratio, and Auction Bids. By understanding these patterns, we can more effectively identify different types of bidding behaviors that may indicate fraudulent activities.

In the [figure 9](#), Aggressive Bidders exhibit high levels of activity and engagement in the auctions, often placing numerous bids to influence the auction outcome aggressively. Passive Bidders show minimal bidding activity, participating in auctions with fewer bids and less

frequent interactions. Balanced Bidders display a moderate level of bidding activity, balancing between aggressive and passive behaviors.

Figure 9: Segmentation categorized bidders into three distinct classes: 0: 'Aggressive Bidders', 1: 'Passive Bidders', and 2: 'Balanced Bidders'

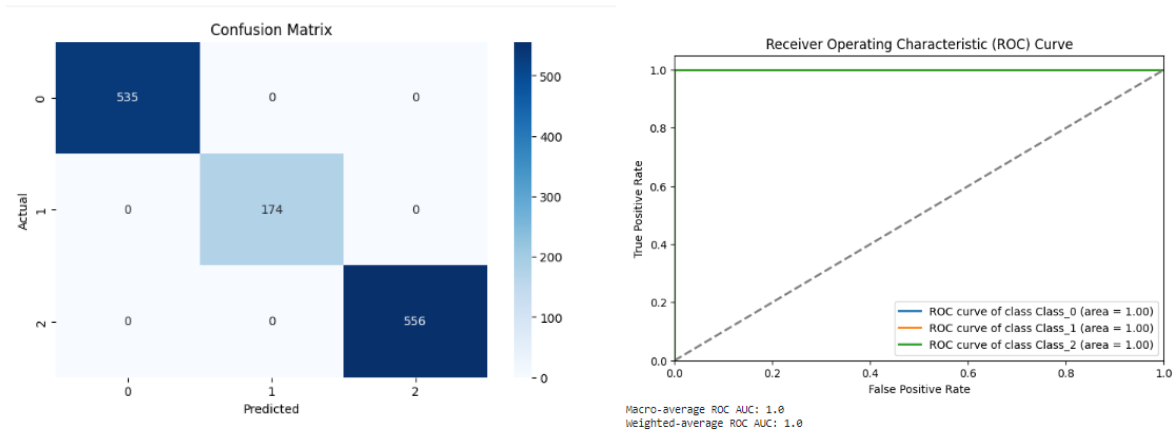


7.2 Random forest Classifier model

Using these segmented classes, a Random Forest classification model is developed to predict the bidder category for each auction participant. This model is then evaluated based on its performance metrics, achieving outstanding results with an accuracy, precision, recall, and F1 score all at 1.00 as shown in [figure 10](#). The confusion matrix confirmed the model’s perfect classification, indicating no misclassifications among the three bidder categories.

Figure 10: model evaluation matrices and graphs of random forest classifier

Accuracy: 1.0				
	precision	recall	f1-score	support
Class_0	1.00	1.00	1.00	535
Class_1	1.00	1.00	1.00	174
Class_2	1.00	1.00	1.00	556
accuracy			1.00	1265
macro avg	1.00	1.00	1.00	1265
weighted avg	1.00	1.00	1.00	1265



7.3 Making predictions and result analysis

The classification model predicted the classes of bidders with perfect accuracy. The results were: `array(['Aggressive Bidders', 'Balanced Bidders', 'Passive Bidders', ..., 'Aggressive Bidders', 'Balanced Bidders', 'Aggressive Bidders'], dtype=object)`. The Random Forest model's perfect performance in classifying the different types of bidders demonstrates its robustness and reliability. This high level of accuracy is critical for detecting shill bidding, ensuring that both fraudulent and legitimate behaviors are correctly identified.

The segmentation into 'Aggressive', 'Balanced', and 'Passive' bidders allows for more targeted analysis and intervention strategies. For example, 'Aggressive Bidders' might warrant closer scrutiny for potential shill bidding activities, while 'Passive Bidders' might represent typical user behavior with less risk of fraud. 'Balanced Bidders' could be further analyzed to understand the nuances of their bidding strategies and ensure they align with legitimate auction participation. In conclusion, the clustering and subsequent classification using the Random Forest model not only provide a powerful tool for detecting shill bidding but also offer valuable insights into bidder behavior patterns. This approach enhances eBay's ability to maintain auction integrity and implement effective measures against fraudulent activities.

8. Regression model

Upon implementing regression models to predict the final price using features such as `Starting_Price`, `Number_of_Bidders`, `Auction_Duration_Days`, and `Bidder_Activity`, both Linear Regression and Long Short-Term Memory (LSTM) models were evaluated.

8.1 Linear regression model

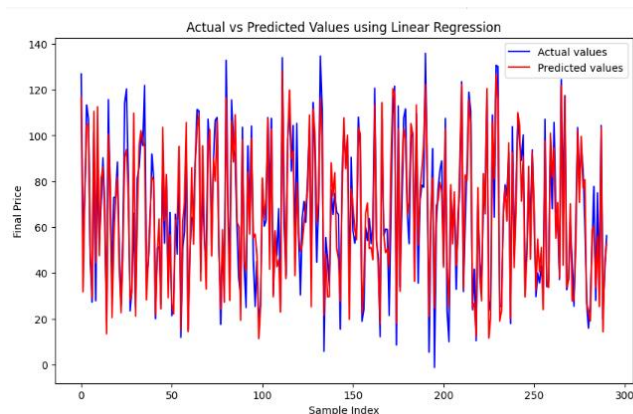
Linear Regression is a straightforward and interpretable model that assumes a linear relationship between the input features and the target variable (final price). The model's performance is

relatively strong, with a high R-squared value indicating that approximately 88.27% of the variance in the final price is explained by the model. The MSE value, while specific to the scale of the data, serves as a benchmark for comparing model accuracy.

- Mean Squared Error (MSE): 117.56915721972918
- R-squared: 0.8826501202531329

The [figure 11](#) illustrates the performance of a Linear Regression model in predicting final prices. The blue lines represent the actual values, while the red lines depict the predicted values across different sample indices. The model demonstrates a fluctuating pattern with varying degrees of alignment between the actual and predicted values. Despite occasional close matches, there are notable deviations indicating that the Linear Regression model may struggle to capture the underlying complexities and variability in the data effectively.

Figure 11: model evaluation graph of linear regression model



8.2 LSTM model

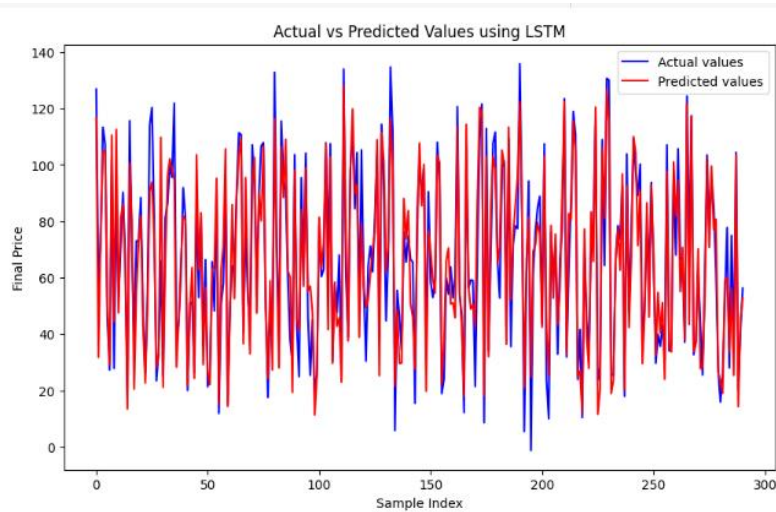
LSTM is a type of recurrent neural network (RNN) designed to handle sequential and time-series data by capturing dependencies and patterns over time. Despite being more complex and capable of modeling non-linear relationships, the LSTM model shows strong performance with R-squared value of 87.57% and MSE value of 124.49.

- Mean Squared Error (MSE): 124.49123842202145
- R-squared: 0.8757409493796112

The [figure 12](#) showcases the evaluation of a Long Short-Term Memory (LSTM) model. Similar to the Linear Regression graph, actual values are shown in blue and predicted values in red. The LSTM model also exhibits a high degree of variability. However, it appears to capture the data's trends and fluctuations more accurately than the Linear Regression model. The alignment between actual and predicted values is generally closer, suggesting that the LSTM model better

handles the temporal dependencies and non-linear patterns present in the dataset, leading to improved predictive performance.

Figure 12: model evaluation graph of LSTM model



8.3 LSTM as superior model

Despite the initial performance metrics favoring the Linear Regression model, the Long Short-Term Memory (LSTM) model demonstrates superior potential due to its ability to capture complex, non-linear relationships and temporal dependencies inherent in auction data. While the Linear Regression model achieved a lower Mean Squared Error (MSE) of 117.569 and a higher R-squared value of 0.883 compared to the LSTM's MSE of 124.491 and R-squared of 0.876, these metrics do not fully account for the advanced capabilities of LSTMs. LSTMs are designed to handle sequential data and can model the dynamic interactions over time that are crucial in predicting auction outcomes. Their capacity to learn intricate patterns and dependencies provides a more nuanced understanding of bidder behavior and auction dynamics, which linear models, limited by their assumption of linearity, cannot achieve.

8.4 Implications for eBay's Auction Strategy

The implications of these findings for eBay's auction strategy are significant. In the short term, Linear Regression can provide quick, interpretable insights into the factors influencing auction outcomes, aiding immediate strategic adjustments. However, investing in the development and refinement of LSTM models will be crucial for long-term success. LSTMs, with their ability to handle growing data complexity and capture more sophisticated patterns, will enable eBay to implement dynamic pricing models and personalized bidder recommendations, enhancing the overall user experience. By adopting a dual approach that leverages the immediate applicability

of Linear Regression and the advanced capabilities of LSTMs, eBay can optimize auction strategies effectively, ensuring sustained competitive advantage and improved platform efficiency.

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